Battery Test

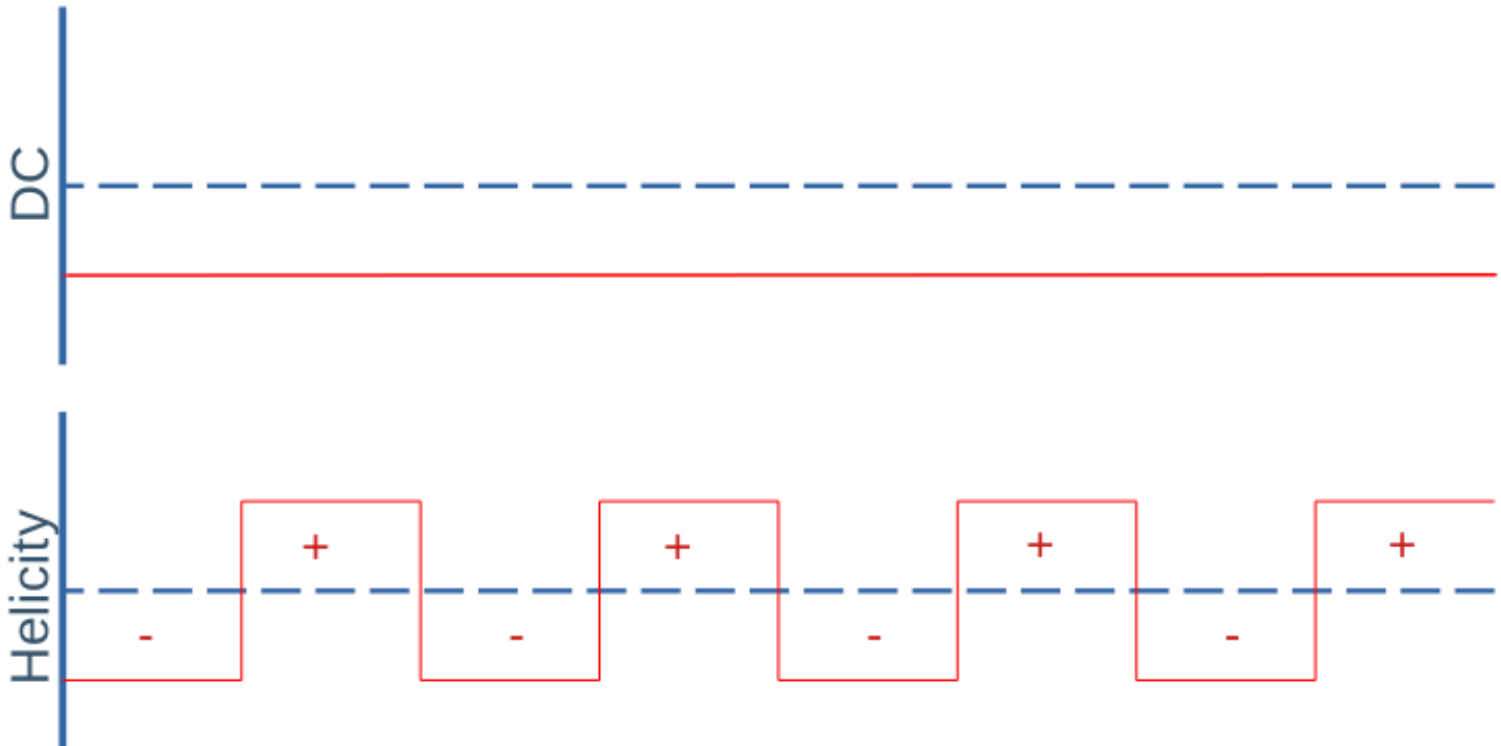
Ryan Conaway, Andrew Gunsch, Addison Arcuri, Joachim Tsakanik

Feb. 24 - 26, 2026

Overview

PVES experiment are sensitive to false asymmetries. It is imperative that we do not "leak" helicity data out of the injector.

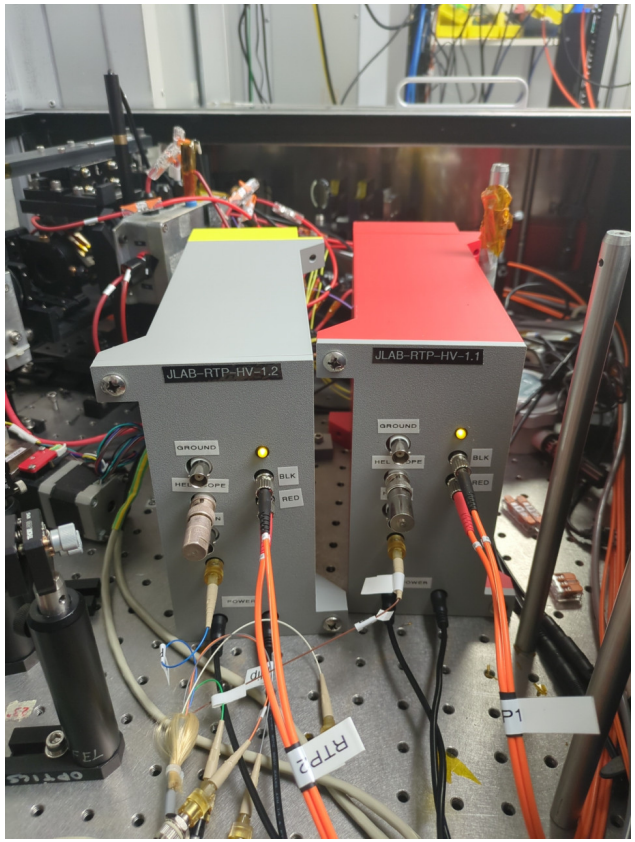
- Goal: Can we pick up helicity information from the laser table, e.g. electronic crosstalk
- Method: Measure DC signals near the helicity source.
  - Null asymmetry $\implies$ not leaking information
  - Finite asymmetry $\implies$ leaking information



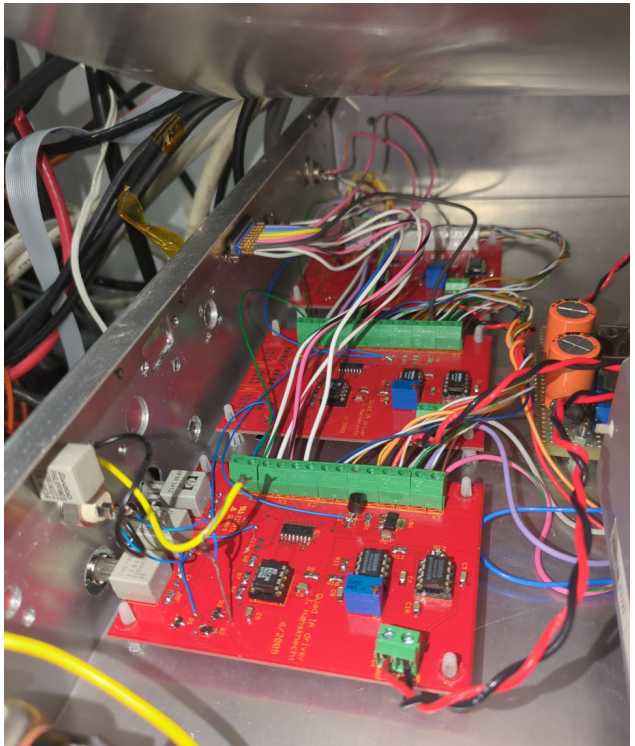
Laser Table Components

We are concerned with crosstalk from two drivers:

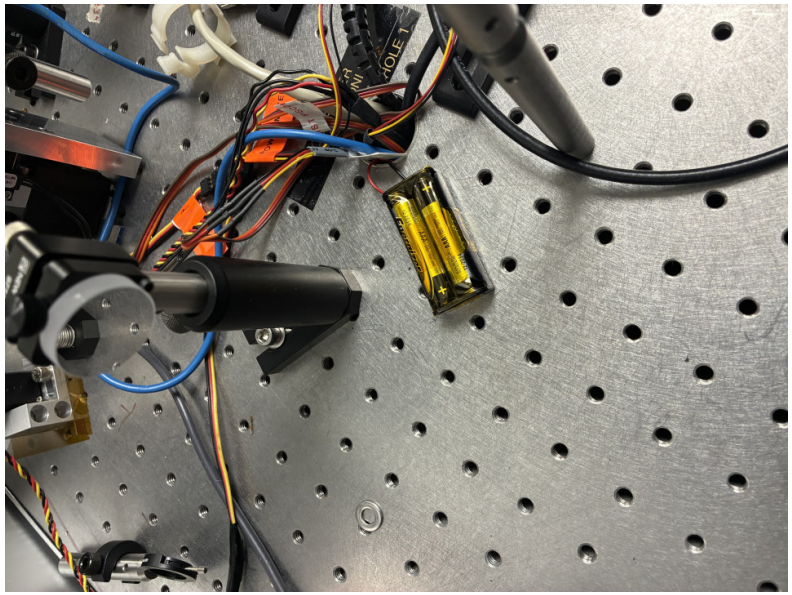
RTP HV Driver



IA Driver



Batteries



Top of laser table, 3.095 V_{dc}

Under laser table, 3.140 V_{dc}

Batteries are isolated but not grounded.

Connected to Inj. ParityDaq

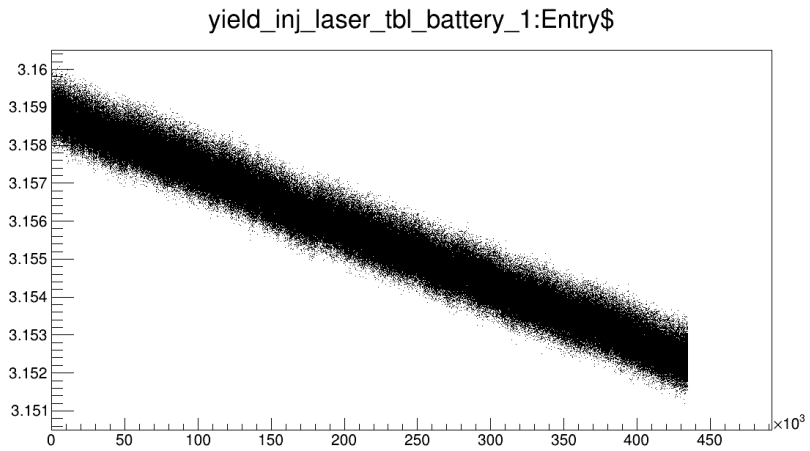
- Under Table (Cable #2) → VQWK , Slot 18, Channel 6
- Top Table (Cable #1) → VQWK , Slot 18, Channel 7
- G0 Bat #1 → VQWK , Slot 18, Channel 8

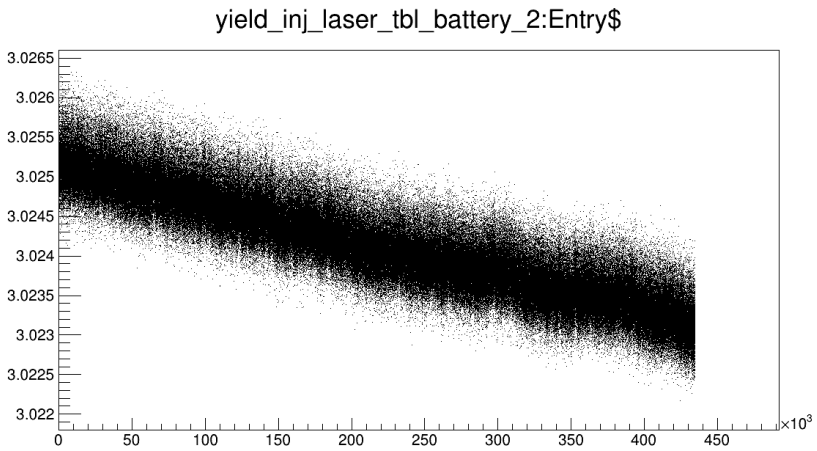
ISB, 3.188 V_{dc}

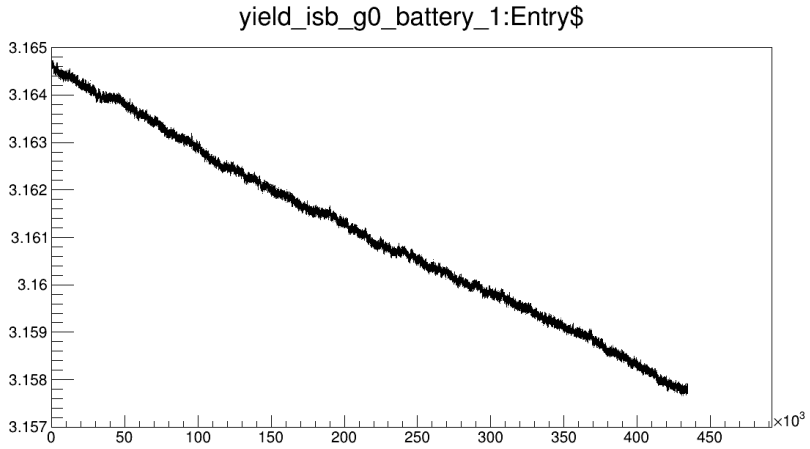
Studies

- RTP HV = OFF, IA = ON
  - Completed, Run 20612: `https://logbooks.jlab.org/entry/4472811`
- RTP HV = ON, IA = ON
  - Completed, Run 20613 and 20614 `https://logbooks.jlab.org/entry/4472950`, `https://logbooks.jlab.org/entry/4472976`
- RTP = ON, IA = OFF
  - Completed, Run 20615 `https://logbooks.jlab.org/entry/4472950`
- RTP = OFF, IA = OFF
  - Completed, Run 20616 `https://logbooks.jlab.org/entry/4473121`
- RTP ON, IA = OFF, RTP PCB grounded
  - Incomplete
- RTP ON, IA = OFF, RTP Isolated, No window delay
  - Incomplete

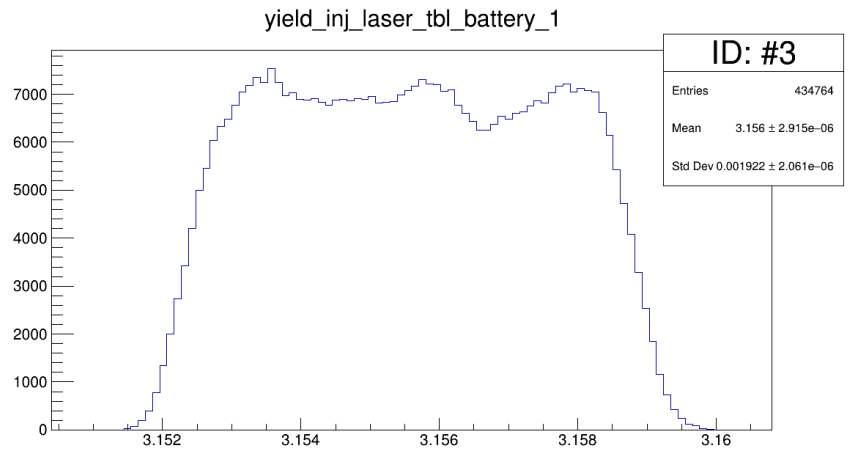
Run 20612: HV = OFF, IA = ON

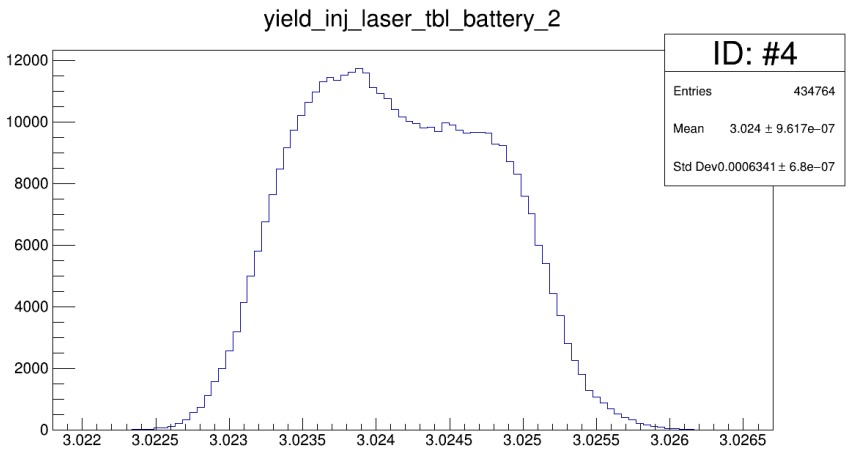


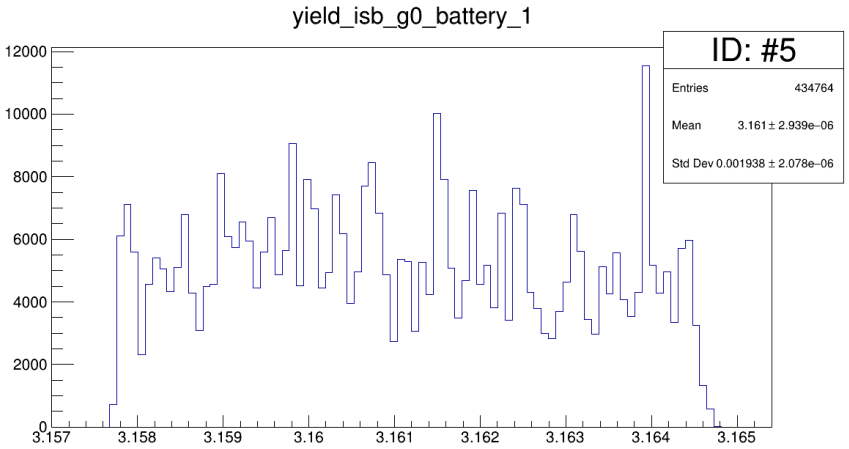




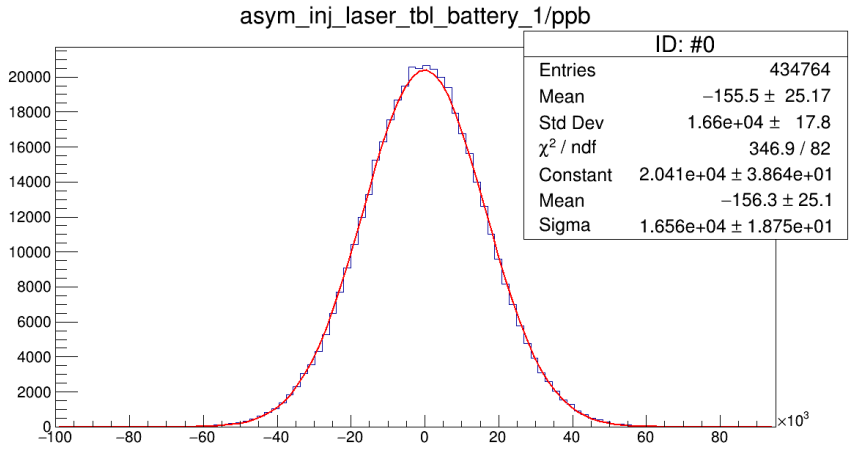
Run 20612: HV = OFF, IA = ON

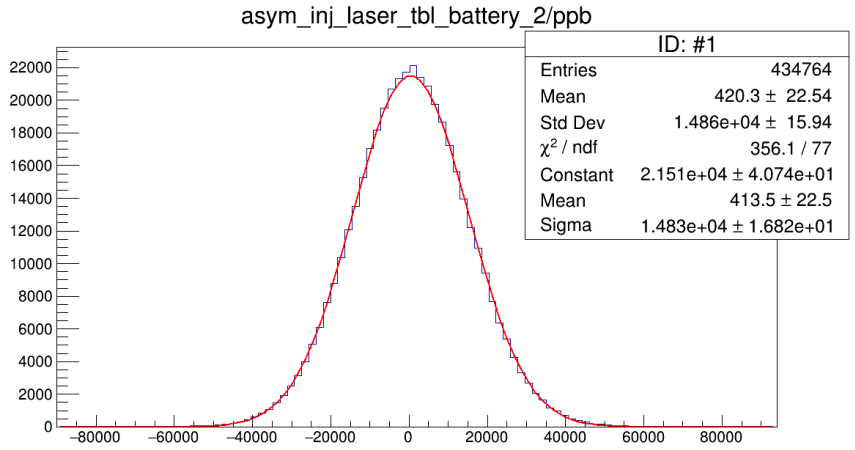


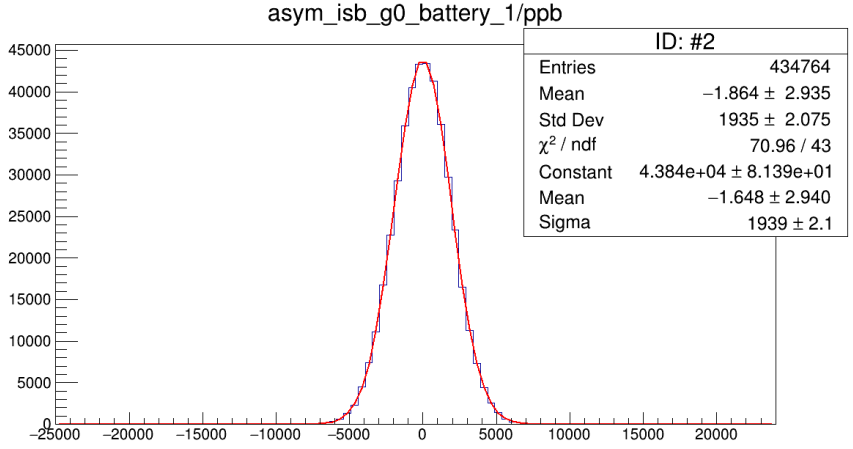




Run 20612: HV = OFF, IA = ON





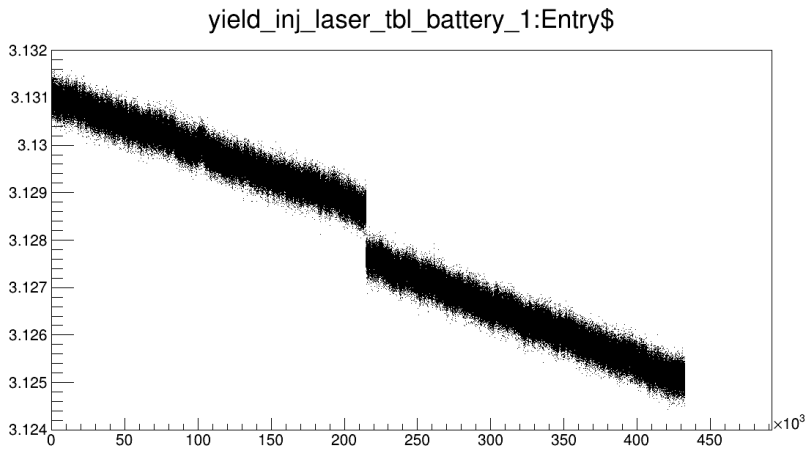


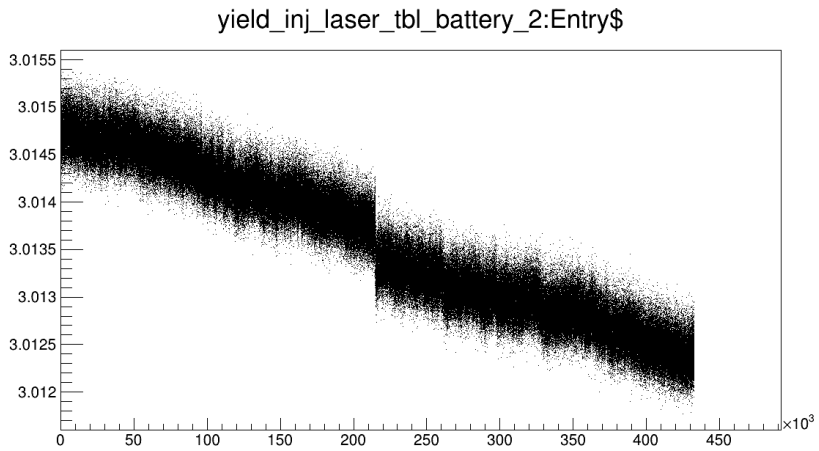
Inj_1: 6.178σ away

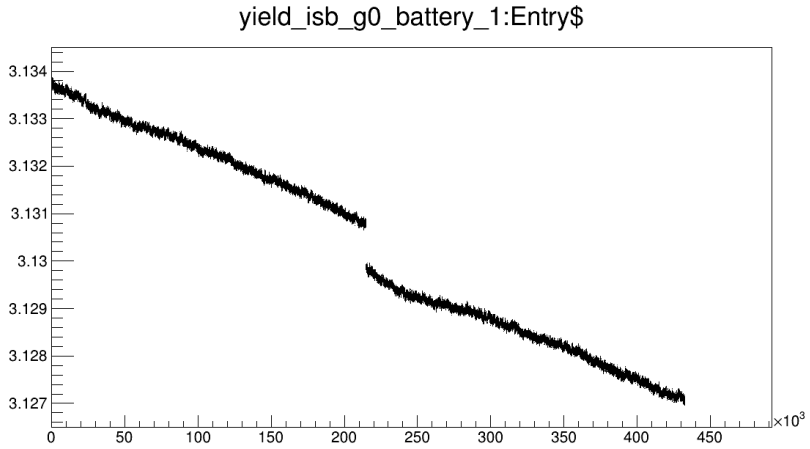
Inj_2: -18.646σ away

ISB: 0.635σ away

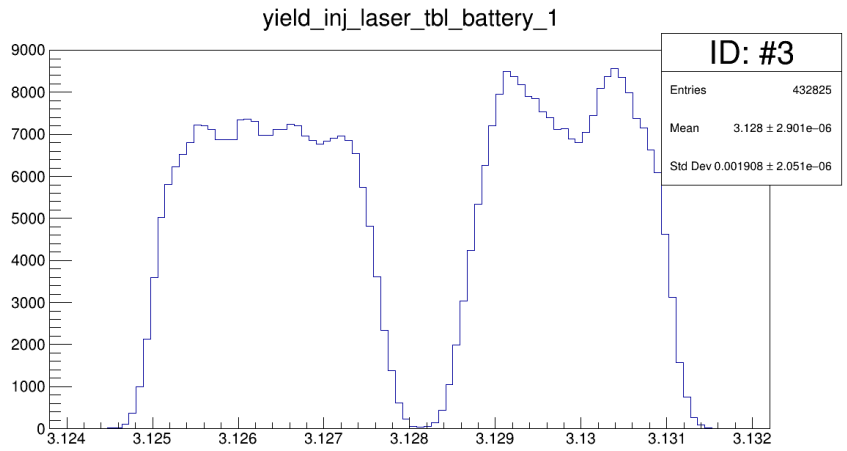
Run 20613 & 201614: HV = ON, IA = ON

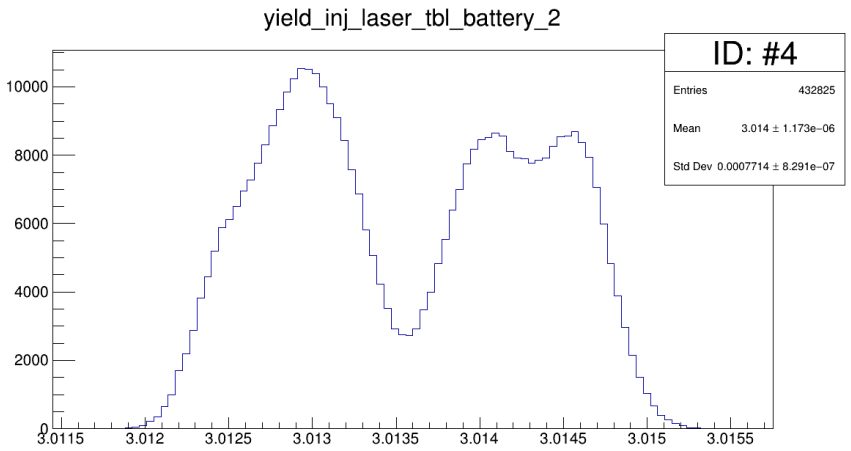


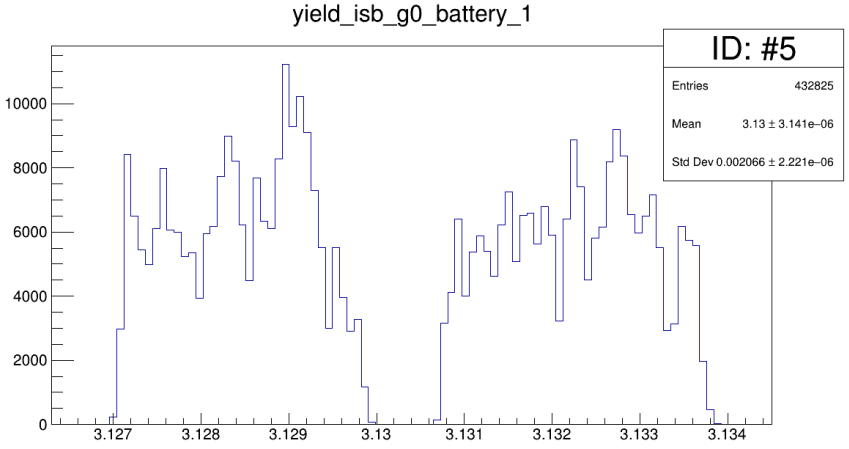




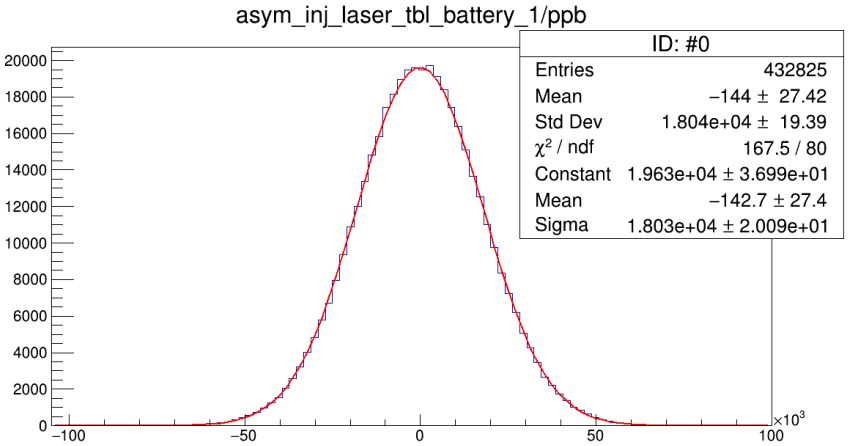
Run 20613 & 201614: HV = ON, IA = ON

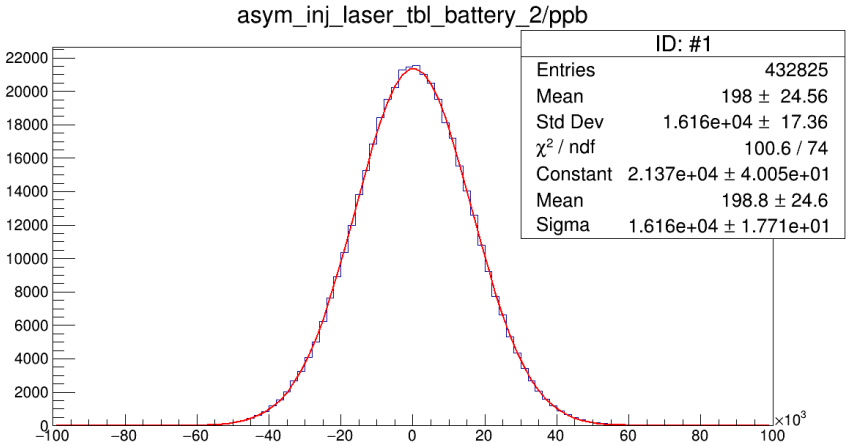


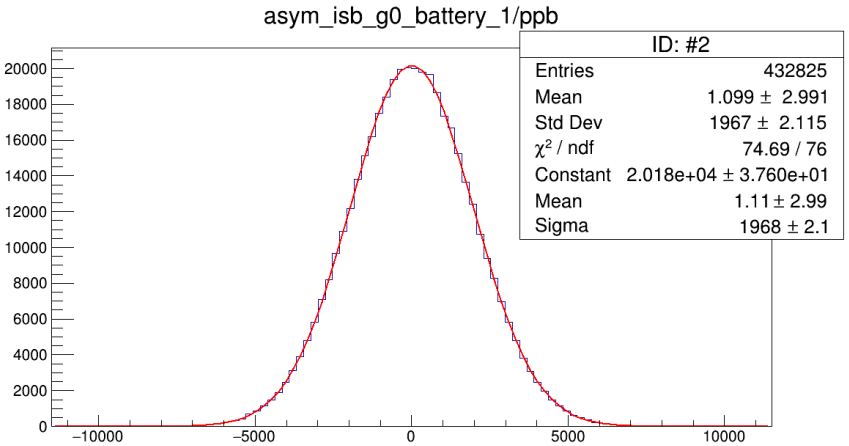




Run 20613 & 201614: HV = ON, IA = ON





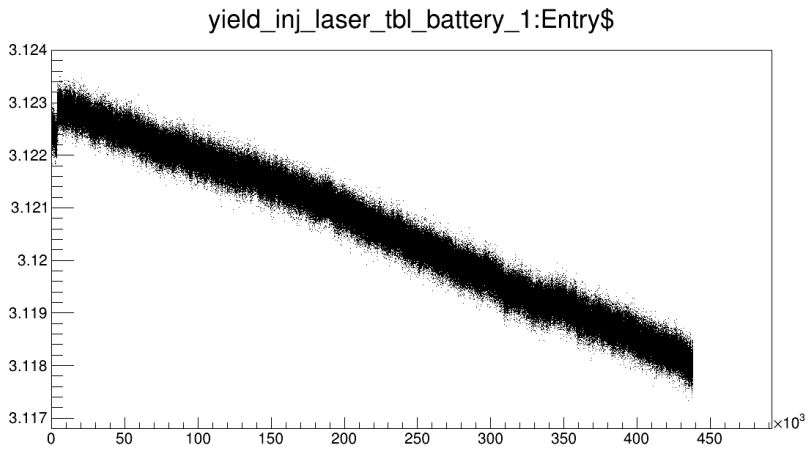


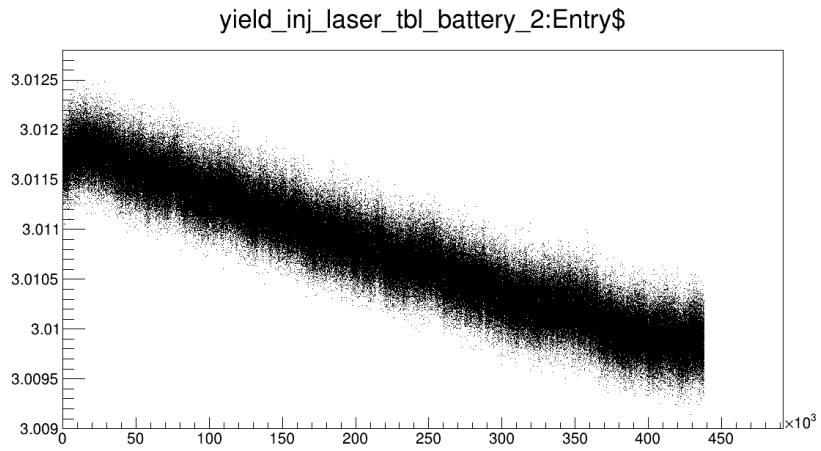
Inj_1: 5.253σ away

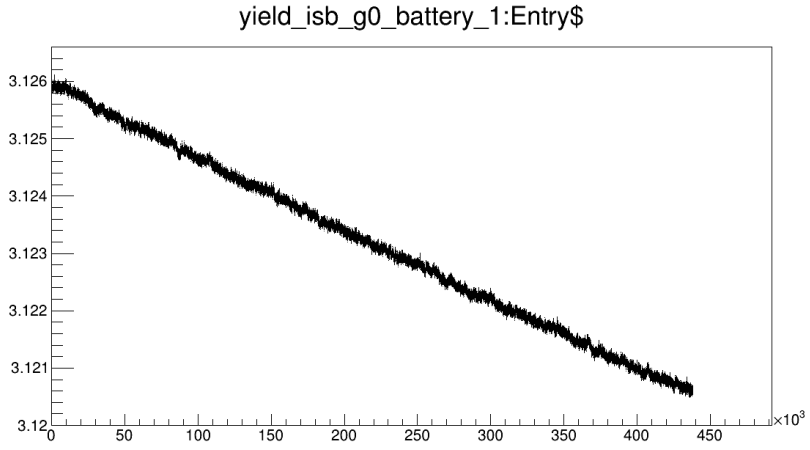
Inj_2: -8.063σ away

ISB: -0.367σ away

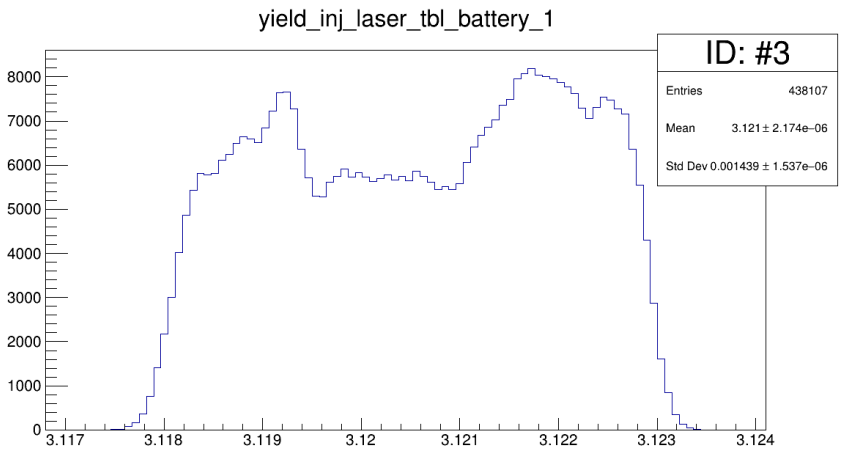
Run 20615: HV = ON, IA = OFF

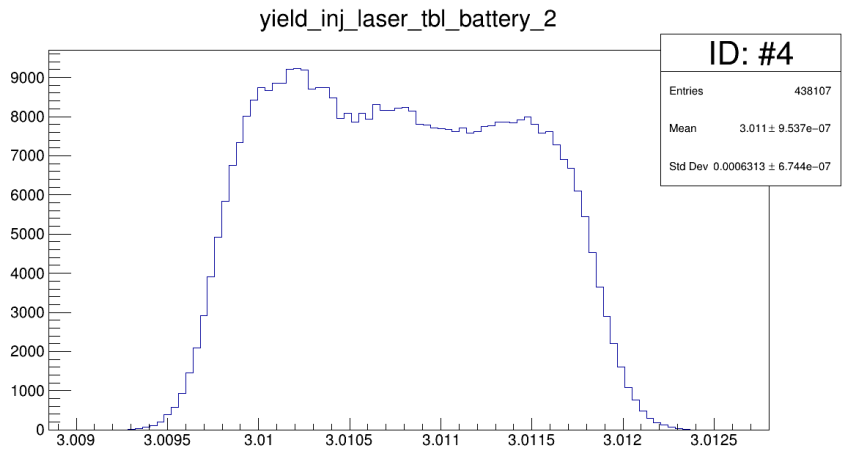


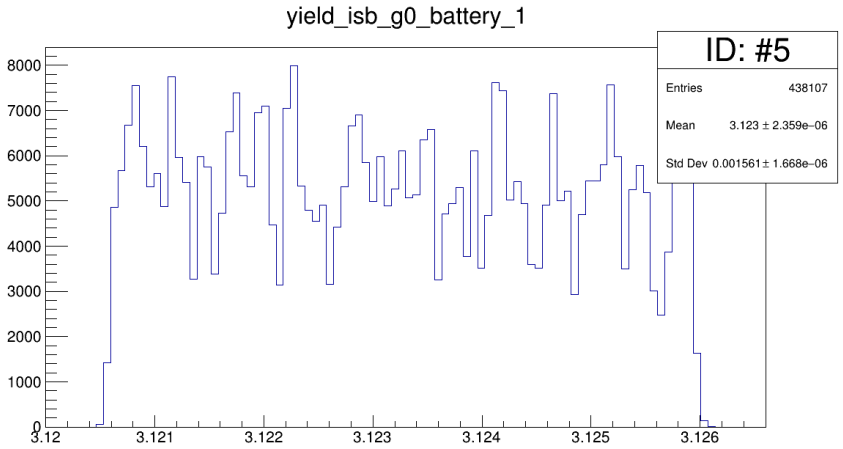




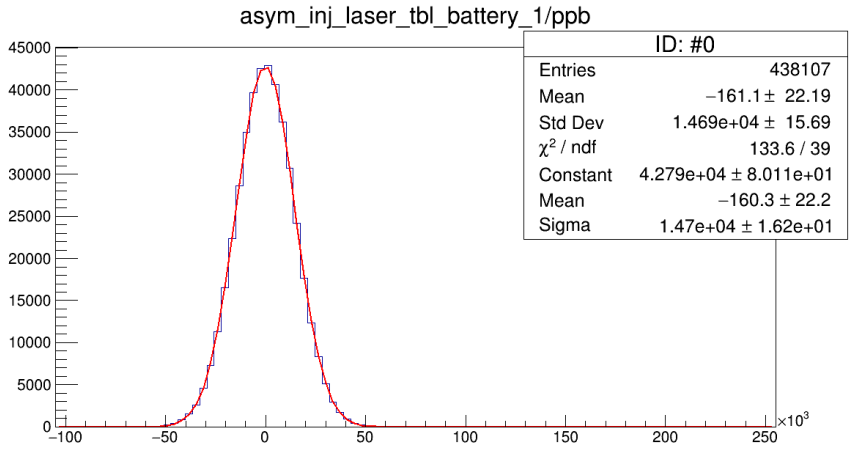
Run 20615: HV = ON, IA = OFF

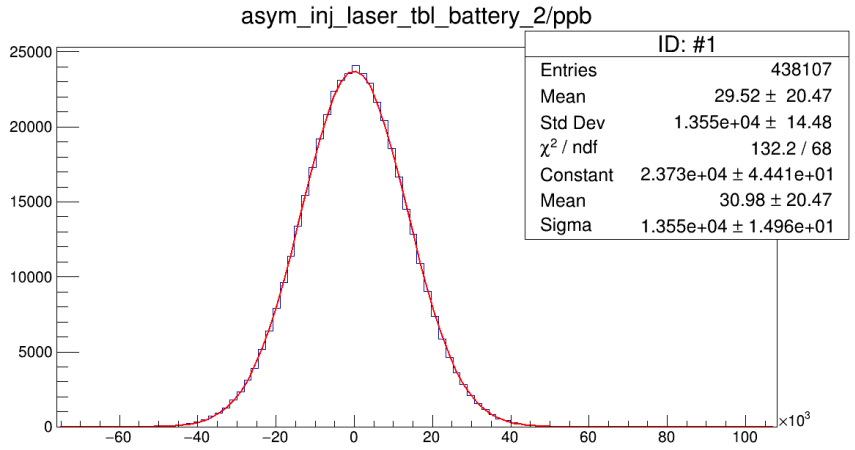


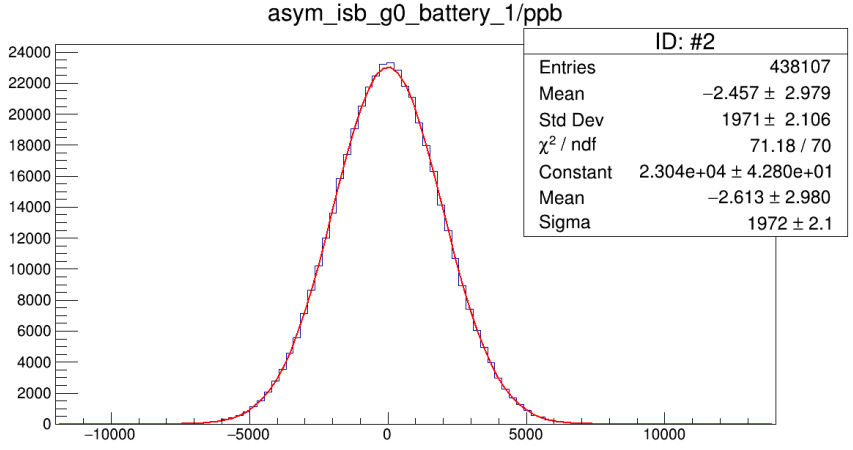




Run 20615: HV = ON, IA = OFF





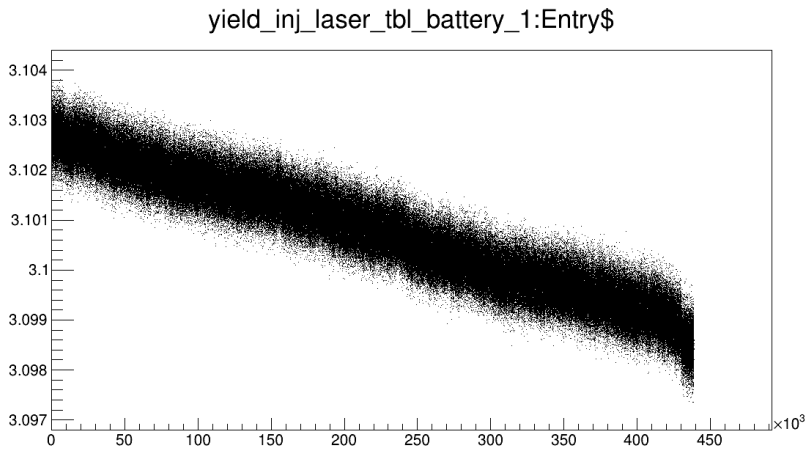


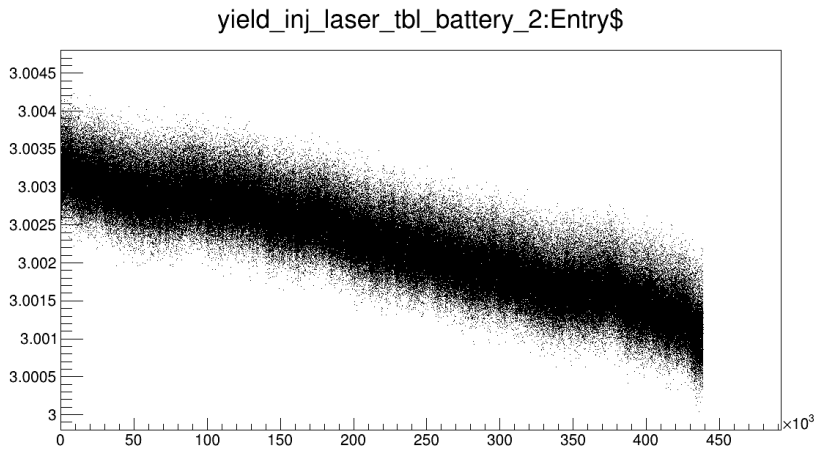
Inj_1: 7.259σ away

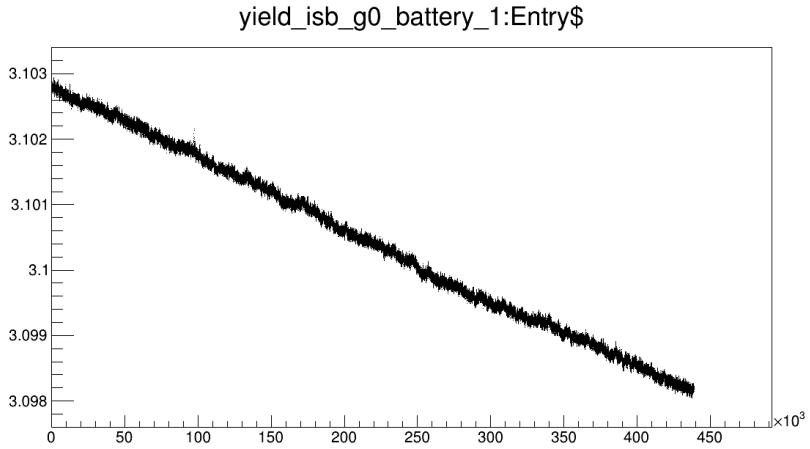
Inj_2: -1.442σ away

ISB: 0.8249σ away

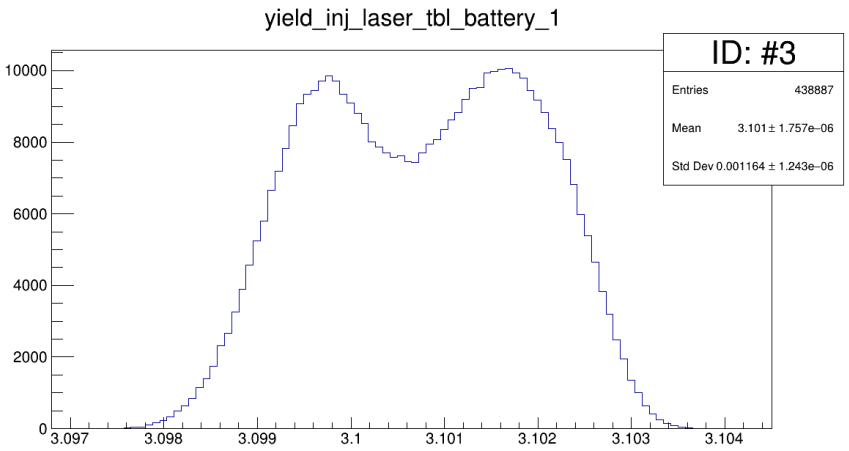
Run 20616: HV = OFF, IA = OFF

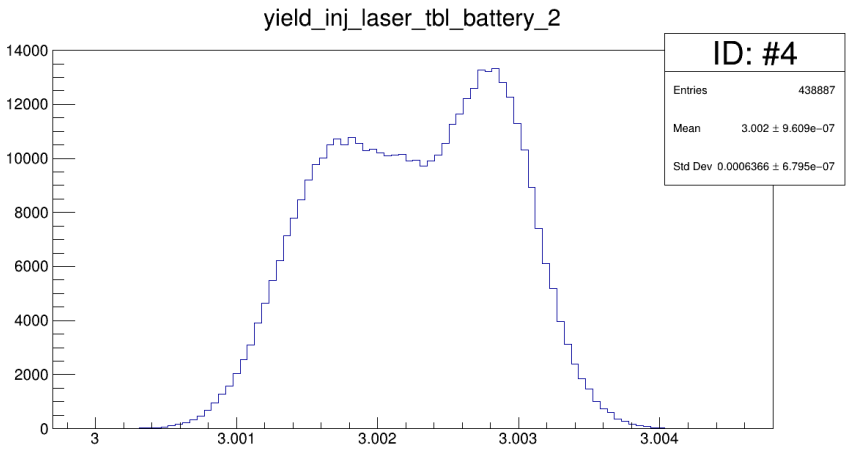


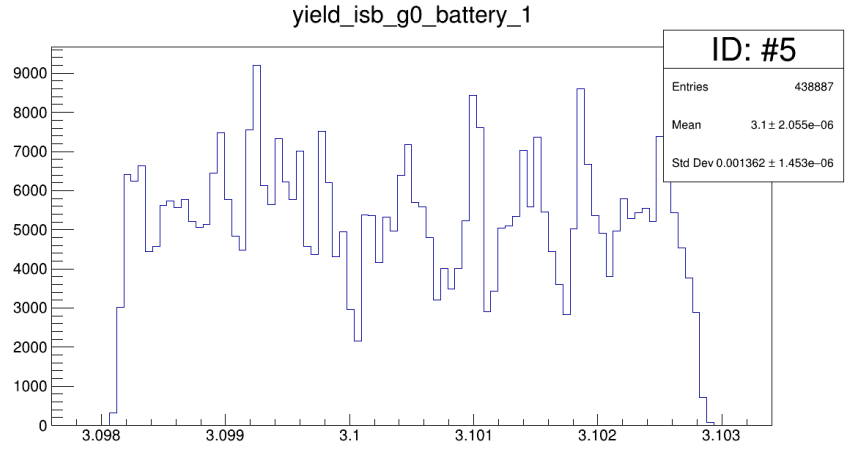




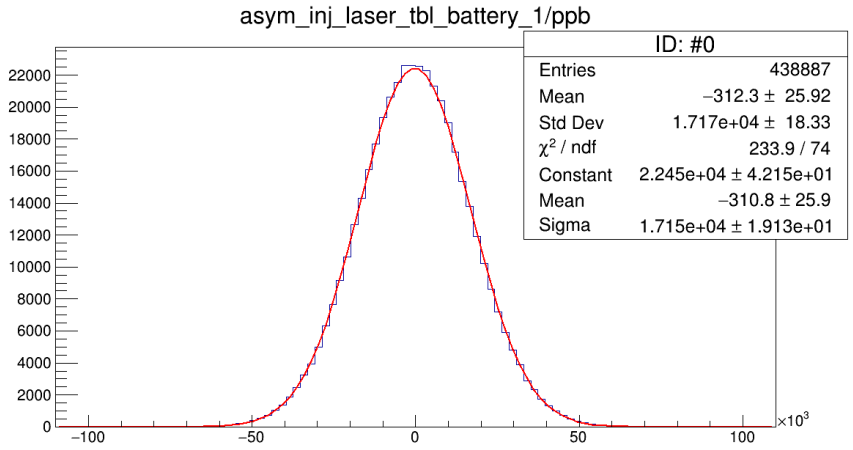
Run 20616: HV = OFF, IA = OFF

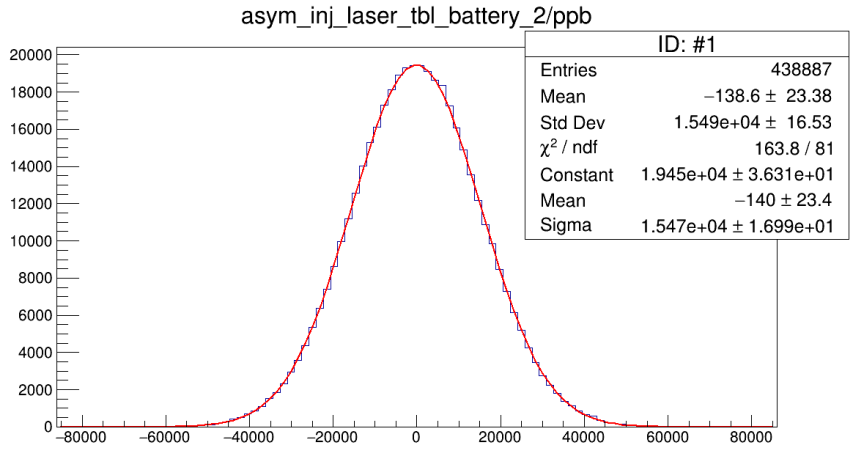


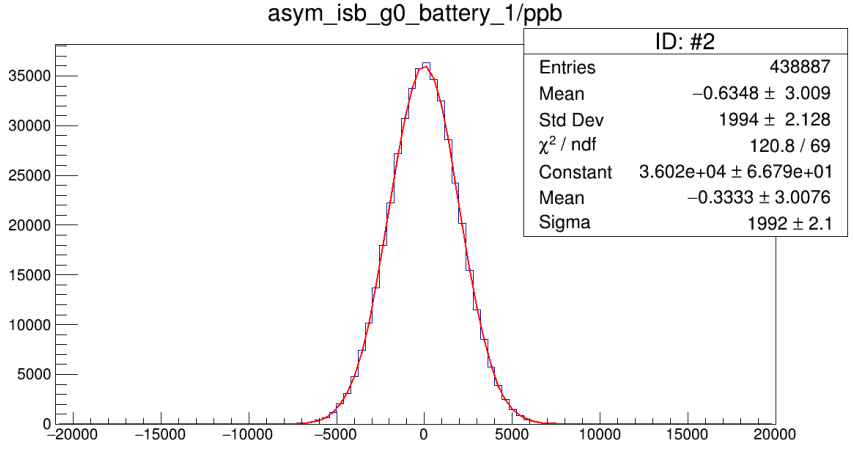




Run 20616: HV = OFF, IA = OFF







Inj_1: 12.049σ away

Inj_2: 5.929σ away

ISB: 0.211σ away

Results

Top of Laser Table

RTP	IA	$\Delta\sigma$
OFF	ON	6.178
ON	ON	5.253
ON	OFF	7.259
OFF	OFF	12.049

Bot of Laser Table

RTP	IA	$\Delta\sigma$
OFF	ON	-18.646
ON	ON	-8.063
ON	OFF	-1.442
OFF	OFF	5.929

ISB

RTP	IA	$\Delta\sigma$
OFF	ON	0.635
ON	ON	-0.367
ON	OFF	0.8249
OFF	OFF	0.211